



# **XpertTeakLite-II<sup>TM</sup>** **Dev A0 Chip**

## **Architecture Specification**

**Rev 1.0.1**

**Dec 2006**



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# 1 Introduction

## 1.1 Scope

The XpertTeakLite-II™ Dev A0 Chip (XpertTeakLite-IIDev) is targeted to serve as a software development platform and a hardware prototyping platform embedding the XpertTeakLite-II DSP platform. The system surrounding the XpertTeakLite-II is minimal, providing a basic interface to a hardware evaluation system.

**Note:** *This XpertTeakLite-IIDev specification specifically conforms to the XpertTeakLite-II DSP Core Architecture Specification V1.2.0 noted below.*

## 1.2 Audience

This document is targeted to the following audiences:

- System (board) designers embedding the XpertTeakLite-IIDev in a system
- VLSI designers using the XpertTeakLite-II IP core
- DSP software programmers using the XpertTeakLite-IIDev

**Note:** *A familiarity with the XpertTeakLite-II platform and the AMBA AHB bus protocol is a prerequisite for this reading.*

## 1.3 Applicable Documents

The following documents relate to the XpertTeakLite-IIDev Chip specification:

- *XpertTeakLite-II Architecture Specification Rev 1.2.0*  
**Note:** *This XpertTeakLite-IIDev Specification specifically conforms to the XpertTeakLite-II DSP Core Architecture Specification version noted above.*
- *CEVA-TeakLite-II Architecture Specification*
- *AMBA AHB Bus Specification (Available online from ARM website)*
- *SOLO PLL Datasheets*

## 1.4 Features List

- Embedded XpertTeakLite-II platform
- XpertTeakLite-II On-chip memory model includes:
  - o 32 KW of tightly coupled data memory
  - o 8 KW Cache subsystem
  - o 32 KW L2 SRAM
- On-chip PLL:
  - o software programmable
  - o PLL bypass option
  - o PLL clock divider
- Interfaces:
  - o TDM Rx and TDM Tx port
  - o AHB master port (L2 memory subsystem)
  - o FLASH Memory Interface with asynchronous burst mode support
  - o MAC-PHY Interface – MII/RMII
  - o PIU (Processor Interface Unit) external APB master port
  - o APB extension port – allows an easy hook-on of additional APB peripherals (EAPB Bridge)
  - o GPIO port
  - o Timers I/F
  - o ICU I/F – supports external interrupt sources
  - o JTAG interface for emulation support
- Stand-alone boot support
- All data ports (AHB, APB, FLASH I/F, TDM and MAC) are bi-directional for pin count reduction
- Bus monitor module – allows analysis of bus activity
  - o AHB access rate counter
  - o Slave refuse rate per master
  - o Indicates Cache miss rate
  - o X/Y double access event counter

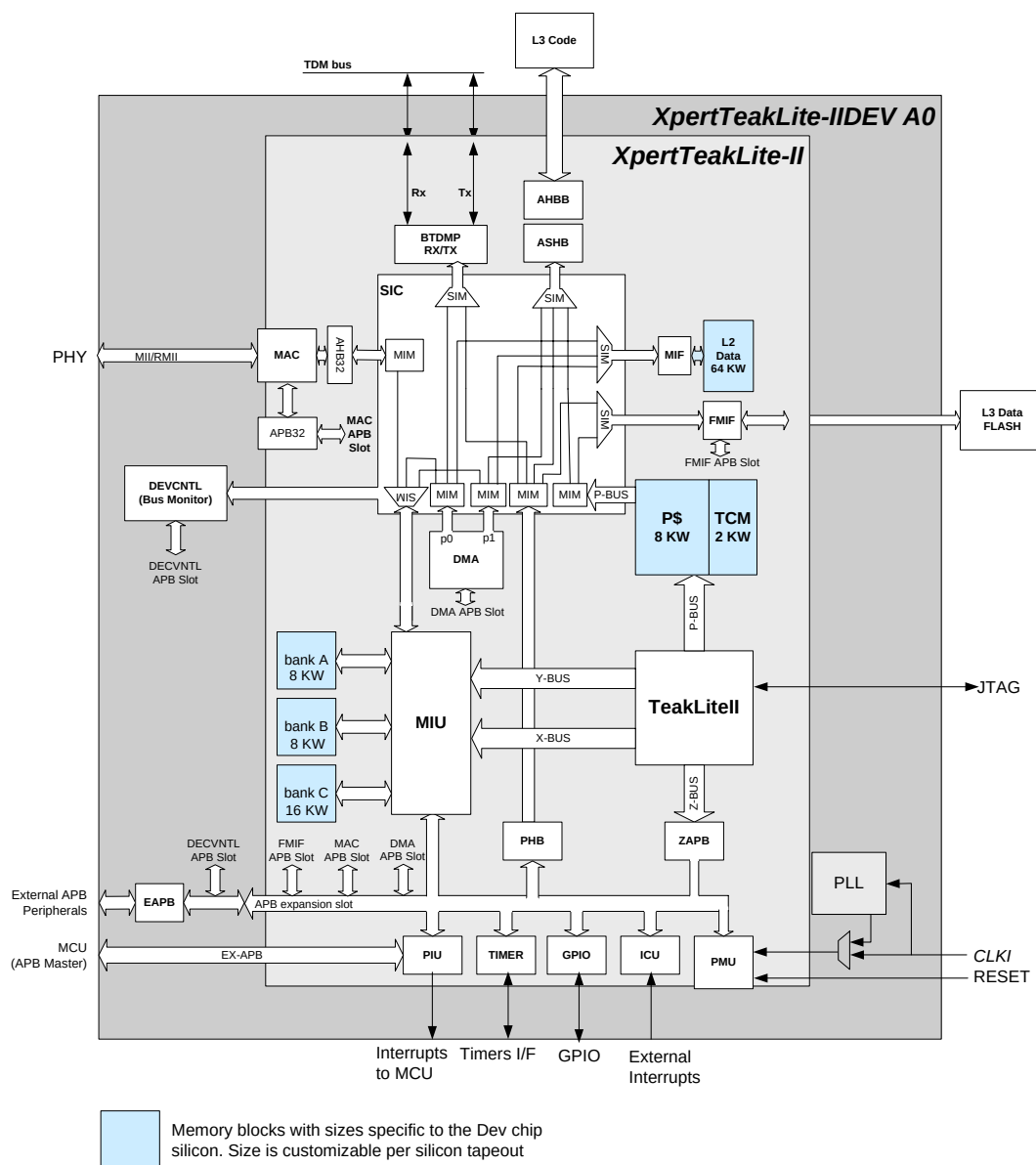
## 1.5 Conventions

The following terms are used frequently throughout this book, and are explained here for ease of reference.

- **CEVA-TeakLite-II** – The CEVA-TeakLite-II DSP core embedded in the XpertTeakLite-II
- **XpertTeakLite-II** – A DSP platform IP core embedded in the development chip
- **AHB** – The AMBA AHB-Lite standard
- **APB** – The Advance Peripheral Bus

## 2 Block Overview

### 2.1 Block Diagram



**Figure 2-1: XpertTeakLite-II Dev Chip Block Diagram**

## 2.2 Blocks Summary

### 2.2.1 XpertTeakLite-II

XpertTeakLite-II is a fully integrated, general-purpose DSP platform that serves a variety of DSP applications. It is equipped with a selected cluster of advanced peripherals and glue-less interfaces. As its backbone, the XpertTeakLite-II offers a powerful and highly efficient, fixed-point CEVA-TeakLite-II DSP and a DMA engine with 3 channels. The DSP, DMA and peripherals are interconnected over a multi-layered AHB-Lite bus architecture. XpertTeakLite-II peripherals include:

- Program Memory Subsystem (PMSS) with
  - o 8 KW direct-map cache
  - o 2 KW Tightly-Coupled Memory (TCM) SRAM
- DMA with 3 channels targeted for:
  - o Data transfer (memory to memory)
  - o TDM TX traffic
  - o TDM RX traffic
- Ethernet MAC 100/10 Unit with MII/RMII interface
- 32 bit AHB Master interface port
- L2 data SRAM of 64 KW
- Memory Interface Unit for CEVA-TeakLite-II fast data RAM with
  - o 8 KW bank A memory
  - o 8 KW bank B memory
  - o 16 KW bank C memory
- TDM receive and transmit channel
- Interrupt Control Unit (ICU)
- Power Management Unit (PMU)
- Processor Interface Unit (PIU)
- General-purpose I/Os (GPIOs)
- Dual timers

### **2.2.2 On-Chip PLL**

A programmable On-Chip PLL is embedded. The PLL offers a programmable power down mode, programmable frequency configuration and a bypass option. These features may be used for an implementation of various power-save modes such as ‘deep-sleep’ mode in which the PLL is off and the system clock is driven directly from the oscillator. The configuration of the PLL is done via the DEVCNTL module programming model.

### **2.2.3 DEVCNTL Module**

This DEVCNTL module serves two main purposes. First, it provides general housekeeping functions (glue logic) specific for the development chip and second, it provides means of analyzing AHB bus activity with in XpertTeakLite-II. The second function is referred to as bus monitoring.

## 3 Chip Interfaces

There are several interface ports in the XpertTeakLite-II Dev for various types of communication with external devices. These are summarized and illustrated in the following sections.

### 3.1 AHB Master Port

This port allows connecting the XpertTeakLite-II Dev as a master over an external AHB-Lite bridge. XpertTeakLite-II devices that can initiate bus requests via this port are: PMSS (CEVA-TeakLite-II code), PHB (CEVA-TeakLite-II Z data) and DMA.

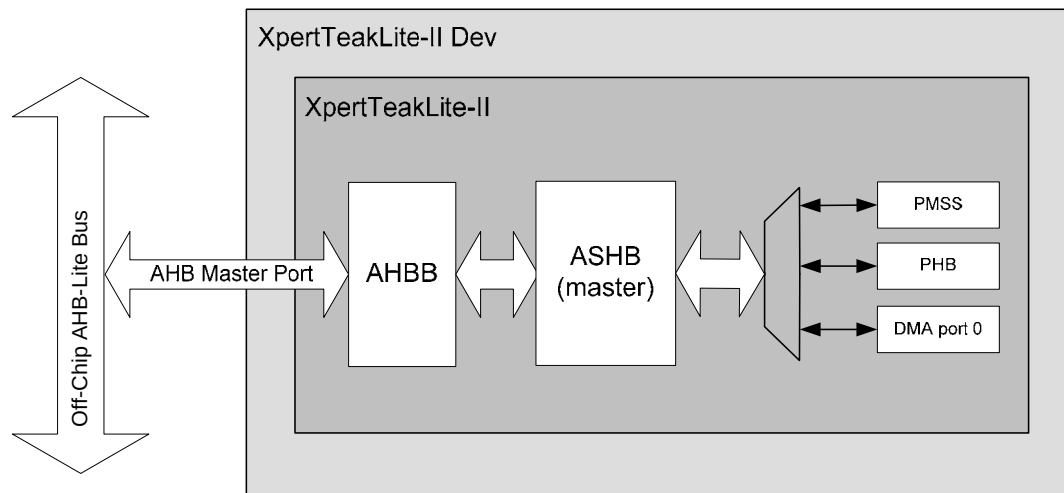


Figure 3-1: AHB Master Port Data Path

#### 3.1.1 Floating the ASHB Master Bus

It is possible for the off chip logic to enforce a high-z state on the master bridge outputs regardless of the state of the output or the bridge. This feature is provided to support a basic arbitration mechanism on an evaluation board. This is done using the input pin **HZ\_ENFORCE**.

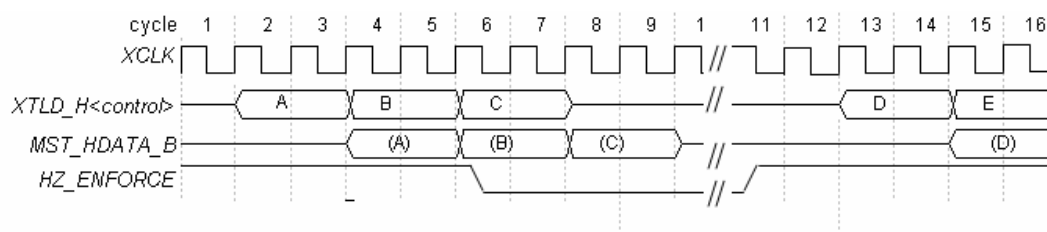
The following ports obtain a high-z state when the **MASTER\_HZ\_ENFORCE** input is held high:

- **XTLD\_HADDR\_U**
- **XTLD\_HBURST\_U**
- **XTLD\_HSIZE\_U**
- **XTLD\_HTRANS\_U**

- **XTLD\_HWRITE\_U**
- **XTLD\_HPROT\_U**
- **XTLD\_HLOCK\_U**

In addition **MST\_HDATA\_B** direction is forced as input

**HZ\_ENFORCE** timing is shown in the diagram below:



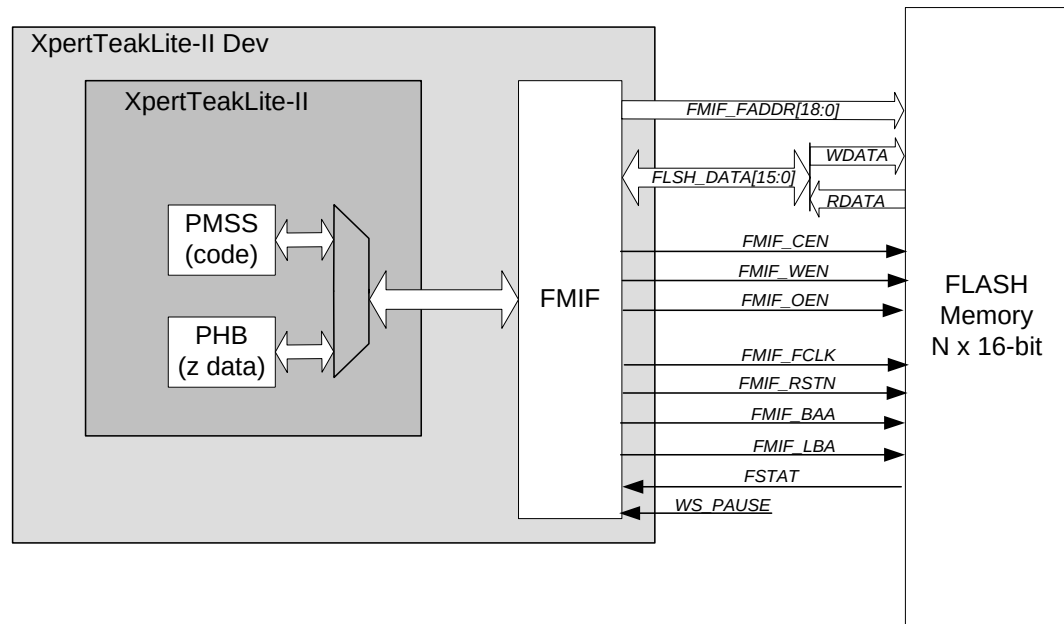
| Signal Name     | Function                                   |
|-----------------|--------------------------------------------|
| XCLK            | System clock                               |
| XTLD_H<CONTROL> | AHB address and control signals            |
| MST_HDATA_B     | Bi-directional data bus of AHB master port |
| HZ_ENFORCE      | Enforce high-z indication                  |

**Figure 2: Hi-Z Enforcement On Master Bridge Outputs**

## 3.2 FLASH Memory Interface Port

An asynchronous FLASH memory device may be hooked as a slave of both the PMSS and the Z data space (PHB). This connection is shown in the figure below:

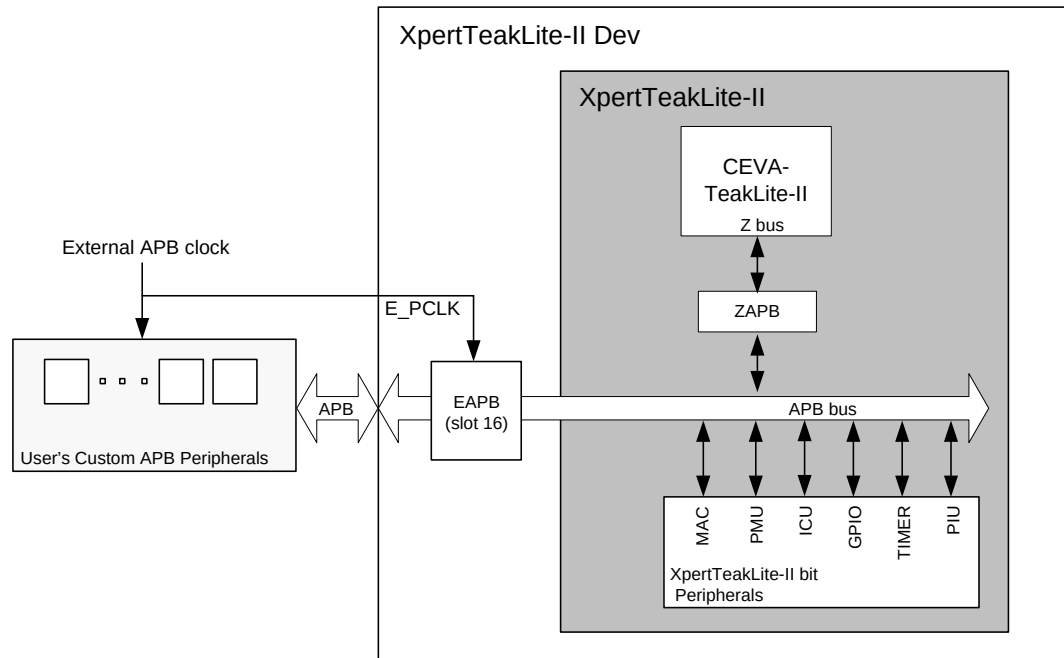




**Figure 3-3: FLASH Memory Connection**

### 3.3 APB Expansion Port

APB custom peripherals may be easily added on to the XpertTeakLite-II Dev chip. These peripherals may be accessed as an integral part of the XpertTeakLite-II Dev peripheral set via a dedicated expansion slot. Note that a single bi-directional data bus is provided in order to reduce the pin count of the chip. The figure below shows how to connect additional APB peripherals to the development chip.



**Figure 3-4: APB Peripheral Expansion**

### 3.4 TDM Port

The XpertTeakLite-II includes 2 TDM port devices, a transmit channel and a receive channel. The DEVCNTL block is responsible for controlling the I/O pad direction for the TDM receive and transmit channels pins.

### 3.5 Interrupt Control Unit Interface

External hardware may use 4 external interrupt input pins to assert CEVA-TeakLite-II interrupt requests. These interrupts should be configured in the Interrupt Control Unit (ICU) programming model. An external device may drive a level-active interrupt or an edge-sensitive interrupt. Refer to the ICU chapter in the XpertTeakLite-II Specification.

## 4 APB Peripherals Address Map

The XpertTeakLite-II APB space facilitates external APB expansion slots for general purpose use. The XpertTeakLite-IIDev utilizes two of the APB expansion slots:

- DEVCNTL APB Slot. This slot hosts the programming model of the DEVCNTL module (PLL configuration and bus-monitoring).
- EAPB APB slot. This slot allows addressing an external (on-board) APB peripheral connected via the external APB port of the development chip

The address bases from the viewpoint of the CEVA-TeakLite-II DSP are listed below:

**Table 4-1: External APB Slots Used in XpertTeakLite-II-Dev**

| APB Slot Name | Slot Number | Address Base (Z space) |
|---------------|-------------|------------------------|
| DEVCNTL_BASE  | 11          | 0x4380                 |
| EAPB_BASE     | 16          | 0x6000                 |

## 5 EAPB Bridge Module

### 5.1 Overview

This module supports connecting external APB peripherals to the XpertTeakLite-II APB subsystem. To accommodate slower clock rates that are expected with off-chip APB system, the bridge slows down the CEVA-TeakLite-II operation when a transaction request is active. The bridge therefore functions as a clock decoupler between the external APB system and the XpertTeakLite-II system.

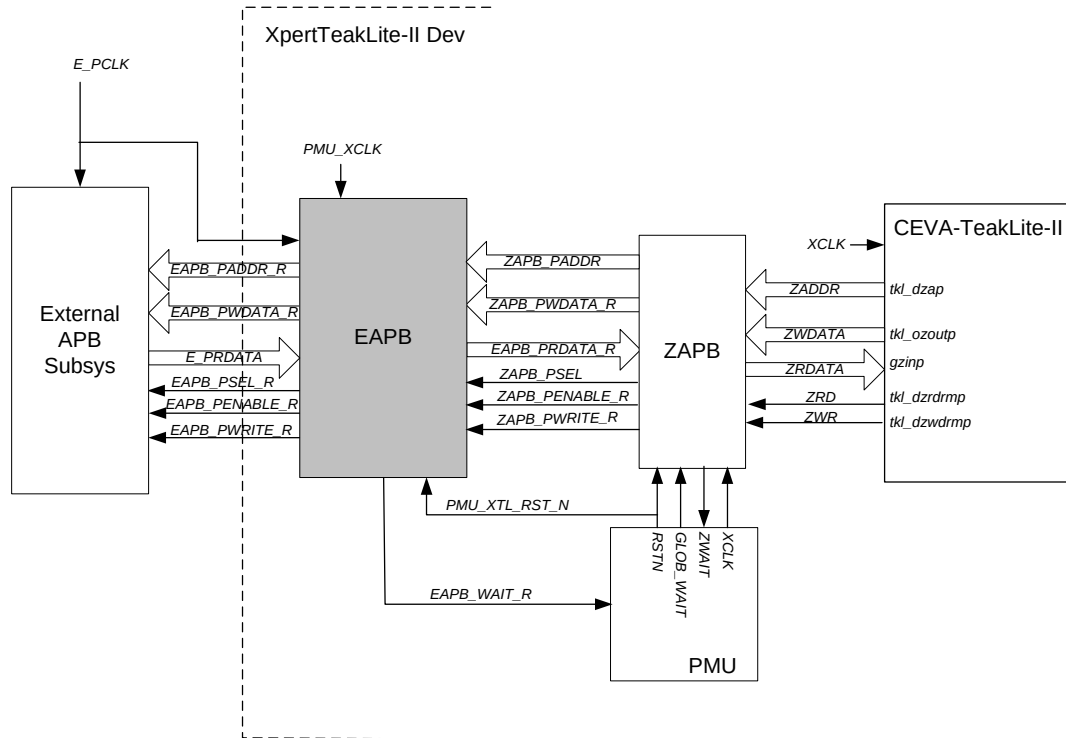
### 5.2 Main Features

The EAPB module has the following main features:

- Bridge functions as an internal APB slave and an external APB master
- Suspends the CEVA-TeakLite-II with wait-states when necessary
- APB slaves may operate on any asynchronous external clock source

## 5.3 Interconnect Diagram

Figure 5-1 illustrates the interconnection of the EAPB device with its surrounding system:



**Figure 5-1: ZAPB Interconnect Diagram**

## 5.4 EAPB Port List

Table 5-1 and Table 5-2 describe the EAPB input and output ports, respectively.

**Table 5-1: EAPB Input Port List**

| Port Name             | Size | Source                   | Description                                                                          |
|-----------------------|------|--------------------------|--------------------------------------------------------------------------------------|
| <b>PMU_XCLK</b>       | 1    | PMU                      | APB subsystem's clock signal                                                         |
| <b>PMU_XTL_RST_N</b>  | 1    | PMU                      | Global reset                                                                         |
| <b>E_PCLK</b>         | 1    | External APB system      | Clock source to the external system used to clock the APB slaves and the EAPB bridge |
| <b>E_PRDATA</b>       | 16   | External APB peripherals | Read data bus from external APB slave(s)                                             |
| <b>ZAPB_PWDATA_R</b>  | 16   | ZAPB                     | Write data bus from ZAPB to EAPB                                                     |
| <b>ZAPB_PWRITE_R</b>  | 1    |                          | Write indication from ZAPB                                                           |
| <b>ZAPB_PENABLE_R</b> | 1    |                          | Chip-select indication from ZAPB                                                     |
| <b>ZAPB_PSEL</b>      | 1    |                          | Chip-select indication from ZAPB                                                     |
| <b>ZAPB_PADDR</b>     | 16   |                          | Address bus from ZAPB to all APB slaves (including EAPB)                             |
| <b>SCANEN</b>         | 1    | TOP                      | Scan enable input                                                                    |
| <b>TESTMODE</b>       | 1    | TOP                      | Test mode input                                                                      |

**Table 5-2: EAPB Output Port List**

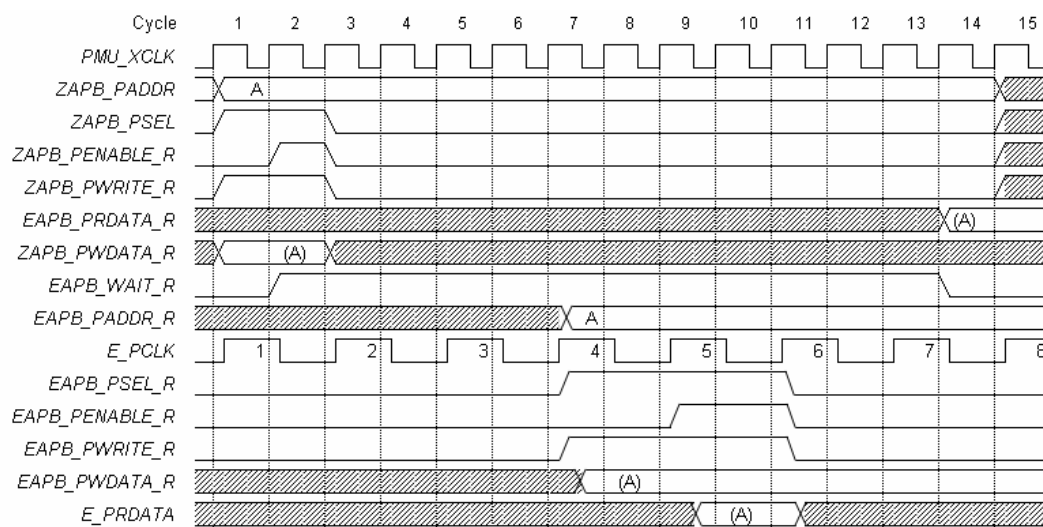
| Port Name             | Size | Destination    | Description                   |
|-----------------------|------|----------------|-------------------------------|
| <b>EAPB_WAIT_R</b>    | 1    | ZAPB           | Wait-state request            |
| <b>EAPB_PRDATA_R</b>  | 16   | ZAPB           | Read data from external slave |
| <b>EAPB_PWRITE_R</b>  | 1    | APB peripheral | Write indication to slave     |
| <b>EAPB_PADDR_R</b>   | 16   |                | Address to external slave     |
| <b>EAPB_PSEL_R</b>    | 1    |                | Select to external slave      |
| <b>EAPB_PENABLE_R</b> | 1    |                | Enable to external slave      |
| <b>EAPB_PWDATA_R</b>  | 16   |                | Write data to external slave  |

## 5.5 EAPB Functionality

When the ZAPB accesses the APB slot dedicated to the EAPB bridge, the bridge responds by issuing a wait-state request to the PMU. The wait request assures that all control and write data lines are kept still until the transaction is complete. Once a request issued by the ZAPB is sensed, the bridge synchronizes the request to the **E\_PCLK** clock domain and issues a standard APB request to the external slave over two **E\_PCLK** cycles. When the data phase of the transaction is complete, an indication is synchronized back to the **PMU\_XCLK** domain that allows releasing the CEVA-TeakLite-II from its wait-state. Read data is synchronized to the **PMU\_XCLK** domain as well.

## 5.6 EAPB Transaction Details

### 5.6.1 Read and Write Example



| Signal Name                                    | Function                       |
|------------------------------------------------|--------------------------------|
| <b>PMU_XCLK</b>                                | System clock                   |
| <b>E_PCLK</b>                                  | External APB clock             |
| <b>EAPB_WAIT_R</b>                             | Wait request from EAPB to ZAPB |
| <b>ZAPB_&lt;ADDR,PSEL,PWRITE,PENABLE&gt;_R</b> | APB controls to the EAPB slot  |
| <b>EAPB_&lt;ADDR,PSEL,PWRITE,PENABLE&gt;_R</b> | External APB controls          |
| <b>EAPB_PWDATA_R</b>                           | External APB device write data |
| <b>EAPB_PRDATA_R</b>                           | Read data returned to ZAPB     |
| <b>E_PRDATA</b>                                | Read data from external slave  |

**Figure 5-2: Read and Write Example**

**Description:**

The request is issued by the ZAPB in the **PMU\_XCLK** clock domain on cycle **PMU\_XCLK(1)**. On the next **PMU\_XCLK** cycle (**PMU\_XCLK(3)**), **EAPB\_WAIT\_R** is asserted in order to stall the transaction initiator, CEVA-TeakLite-II. The address, control and write data from (**ZAPB\_PADDR**, **ZAPB\_PSEL**, **ZAPB\_PWDATA\_R** respectively) are latched internally once by **PMU\_XCLK(3)** when **ZAPB\_PENABLE\_R** is set high. At this stage, a handshake is carried over between the two clock domains. The address is handed over to the **E\_PCLK** side on the rising edge of **E\_PCLK(7)**. Next, a standard APB transfer is carried over cycles **E\_PCLK(7-8)**. In case of read, the read data is internally sampled with **E\_PCLK** on cycle **E\_PCLK(11)**. The valid read data is issued over **EAPB\_PRDATA\_R** on cycle **PMU\_XCLK(14)**. At the same time, **EAPB\_WAIT\_R** is released and the read data is taken by the ZAPB on the **PMU\_XCLK(15)** clock rise.



## 6 DEVCNTL – Glue Logic and Bus Monitoring

### 6.1 Overview

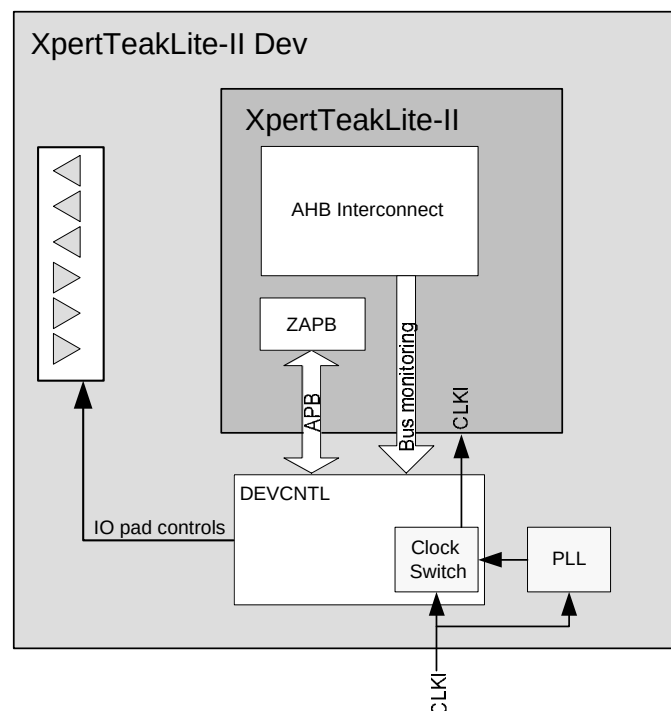
The DEVCNTL module is an APB peripheral with two main purposes:

- Bus monitoring function
- General Purpose Glue Logic specific for the XpertTeakLite-II Dev

With respect to bus monitoring, this feature enables visibility of the traffic load on the AHB matrix and internal arbitration statistics. Bottlenecks caused by real-time conflicts over shared resources may be analyzed off-line by reading statistics from special event counters that snoop the AHB matrix state on every cycle.

The general purpose glue logic facilitates general housekeeping tasks like bi-directional I/O pad control for all bonded out data lines, configuring and multiplexing dual-function I/O-s and PLL clock bypassing.

The diagram below illustrates the connectivity of the DEVCTL module within the system.



**Figure 6-1: DEVCNTL Module Connection**

### **6.1.1 Main Features**

- Includes 26 bus monitor event-counters for analyzing real-time bus events
- APB interface for extracting the bus monitoring data and configuring glue logic functions
- General housekeeping functions:
  - o Output enable signaling for all bidirectional data buses
  - o Boot strap control – selects the 32-bit boot target vector at reset from external data buses
  - o PLL Configuration via APB programmer model
  - o Clock selection – Selects between PLL and direct oscillator clock as system main clock input
  - o Dual-function I/O selection and multiplexing

## 6.2 DEVCNLT Port List

**Table 6-1: DEVCNLT Input Port List**

| Input Port Name          | Size  | Description                                                    |
|--------------------------|-------|----------------------------------------------------------------|
| <i>pmu_xclk</i>          | 1     | Peripherals clock                                              |
| <i>ex_rst_n</i>          | 1     | External reset signal (clk switch input)                       |
| <i>xtl_rst_n</i>         | 1     | XpertTeakLite-II system reset (active-low)                     |
| <i>testmodep</i>         | 1     | External test mode indication                                  |
| <i>scanen</i>            | 1     | Scan enable indication                                         |
| <i>xtl_sco</i>           | 54    | Scan chains output from XpertTeakLite-II                       |
| <i>s_hrdata</i>          | 32    | Scan-in through AHB Slave read data bus                        |
| <i>frdata</i>            | 16    | Scan-in through Flash Slave read data bus                      |
| <i>apb_prdata</i>        | 6     | Scan-in through APB Slave read data bus                        |
| <i>sci</i>               | 5     | Scan-in through APB Slave read data bus – for DevCtrl scanning |
| <i>xtl_glob_wait</i>     | 1     | PMU global wait signal to TeakLite-II                          |
| <i>xtl_haddr_r</i>       | 32    | Scan-out through AHB slave address bus                         |
| <i>xtl_fmif_faddr_r</i>  | 19    | Scan-out through FLASH write data bus                          |
| <i>xtl_hsize_r</i>       | 3     | Scan-out through AHB slave transaction size bus                |
| <i>eapb_paddr_r</i>      | 9     | EAPB address bus for DevCtrl Scan mux                          |
| <i>osc_clk</i>           | 1     | Oscillator clock                                               |
| <i>pll_clk</i>           | 1     | PLL clock                                                      |
| <i>xtl_psel</i>          | 1     | APB select signal                                              |
| <i>xtl_penable_r</i>     | 1     | APB enable signal                                              |
| <i>xtl_pwrite_r</i>      | 1     | APB write signal                                               |
| <i>xtl_paddr</i>         | 6     | APB address bus                                                |
| <i>xtl_pwdata_r</i>      | 16    | APB data bus                                                   |
| <i>tkl_trapap</i>        | 1     | trap interrupt indication                                      |
| <i>mimN_xhtrans</i>      | 2 x 5 | MIM #N Transaction mode (N=0:4)                                |
| <i>simN_htrans</i>       | 2 x 5 | SIM #N Transaction mode (N=0:4)                                |
| <i>mimN_in_xhaready</i>  | 1 x 5 | MIM #N AREADY input bus                                        |
| <i>simN_out_xhaready</i> | 5 x 5 | SIM #N AREADY output bus to the different MIMs                 |
| <i>mimN_hready</i>       | 1 x 5 | MIM #N HREADY signal to the master                             |
| <i>s_hclk</i>            | 1     | AHB Slave hclk                                                 |
| <i>xtld_htrans_r</i>     | 2     | AHB transaction type                                           |
| <i>xtl_hwrite_r</i>      | 1     | AHB write transaction                                          |
| <i>s_hready</i>          | 1     | AHB slave HREADY signal                                        |
| <i>eapb_pclk</i>         | 1     | EAPB clock                                                     |
| <i>eapb_psel_r</i>       | 1     | EAPB select signal                                             |
| <i>eapb_pwrite_r</i>     | 1     | EAPB write signal                                              |

| Input Port Name                        | Size | Description                                                     |
|----------------------------------------|------|-----------------------------------------------------------------|
| <i>ext_pclk</i>                        | 1    | External APB clock (MCU)                                        |
| <i>ext_psel</i>                        | 1    | External APB select signal                                      |
| <i>ext_pwrite</i>                      | 1    | External APB write signal                                       |
| <i>xtl_fmif_pad_oe_r</i>               | 1    | PAD output enable to dev control                                |
| <i>xtl_btdmpr_fsp_oe_n_r</i>           | 1    | Receive Channel, Frame Synchronization Pulse Output Enable      |
| <i>xtl_btdmpr_cntrl_tdmclk_oe_n_cr</i> | 1    | Receive Channel, TDMCLK Output Enable                           |
| <i>xtl_btdmpt_plmt_tdout_oe_n_cr</i>   | 1    | Transmit Channel, Serial Transmit DATA OUT Output Enable        |
| <i>xtl_btdmpt_fsp_oe_n_r</i>           | 1    | Transmit Channel, Frame Synchronization Pulse Output Enable     |
| <i>xtl_btdmpt_cntrl_tdmclk_oe_n_cr</i> | 1    | Transmit Channel, TDMCLK Output Enable                          |
| <i>ashb_hz_enforce</i>                 | 1    | A high-z state on the master bridge                             |
| <i>xtld_ahb_hrdata</i>                 | 32   | AHB Master data bus. Used as target boot vector (boot strap)    |
| <i>ex_wait</i>                         | 1    | External CEVA-TeakLite-II wait-state request                    |
| <i>buin0p</i>                          | 1    | User input #0                                                   |
| <i>buin1p</i>                          | 1    | User input #1                                                   |
| <i>ext_nmi</i>                         | 1    | External NMI                                                    |
| <i>ext_int_src</i>                     | 3    | External interrupt                                              |
| <i>ext_bp_req</i>                      | 1    | External break point (abort) request                            |
| <i>pll_lock</i>                        | 1    | External PLL lock indication                                    |
| <i>gpio_in</i>                         | 1    | General purpose input used for capture debug data (strap logic) |
| <i>bist_en</i>                         | 1    | BIST mode                                                       |
| <i>xtld_timer0_tout_r</i>              | 1    | Timer_unit 0 interrupt                                          |
| <i>xtld_timer0_pwm_r</i>               | 1    | Timer_unit 0 PWM output signal                                  |
| <i>xtld_timer1_tout_r</i>              | 1    | Timer_unit 1 interrupt                                          |
| <i>xtld_timer1_pwm_r</i>               | 1    | Timer_unit 1 PWM output signal                                  |
| <i>prg_bist_out</i>                    | 4    | Program BIST output                                             |
| <i>otdi</i>                            | 1    | JTAG TDI                                                        |
| <i>otms</i>                            | 1    | JTAG TMS                                                        |
| <i>s_hresp</i>                         | 1    | AHB Slave response                                              |
| <i>ext_paddr</i>                       | 4    | APB address from external APB master host                       |
| <i>xtld_fmif_we_n_r</i>                | 1    | Write enable - active low                                       |
| <i>xtld_fmif_oe_n_r</i>                | 1    | Output enable - active low                                      |
| <i>xtld_fmif_ce_n_r</i>                | 1    | Chip enable - active low                                        |
| <i>xtld_fmif_baa_r</i>                 | 1    | Burst address advance indication                                |
| <i>data_bist_out</i>                   | 4    | Data BIST output                                                |
| <i>ref_clk</i>                         | 1    | MAC – 50MHz reference clock input                               |
| <i>ext_penable</i>                     | 1    | APB enable from external APB master host                        |

| Input Port Name              | Size | Description                              |
|------------------------------|------|------------------------------------------|
| <i>fpause</i>                | 1    | Pause transfer indication                |
| <i>fstat</i>                 | 1    | General purpose status input             |
| <i>col</i>                   | 1    | Collision detect signal from the PHY     |
| <i>crs</i>                   | 1    | Carrier sense signal from the PHY        |
| <i>cxld</i>                  | 4    | Transmit data to the PHY                 |
| <i>xtld_uout0p</i>           | 1    | User output 1                            |
| <i>xtld_uout1p</i>           | 1    | User output 0                            |
| <i>xtld_piu_dsem_int_r</i>   | 1    | Semaphore interrupt to External MCU      |
| <i>xtld_piu_comrep_int_r</i> | 1    | Command read interrupt to External MCU   |
| <i>L2_bist_out</i>           | 4    | L2 BIST output                           |
| <i>link</i>                  | 1    | Status of link pin from the PHY          |
| <i>tx_clk_source</i>         | 1    | MAC transmit clock source from PHY       |
| <i>ext_clktimer0</i>         | 1    | External Clock for TIMER_UNIT0           |
| <i>ext_clktimer1</i>         | 1    | External Clock for TIMER_UNIT1           |
| <i>ext_tdm_rd_in</i>         | 1    | Receive Channel - Serial Receive DATA IN |
| <i>ex_wake</i>               | 1    | External clock recovery request          |
| <i>rx_er</i>                 | 1    | Receive error signal from the PHY        |

Table 6-2: DEVCNTRL Output Port List

| Output Port Name                   | Size | Description                                                                      |
|------------------------------------|------|----------------------------------------------------------------------------------|
| <i>dctl_rst_n</i>                  | 1    | XpertTeakLite-II reset signal (active low)                                       |
| <i>dctl_prdata_cr</i>              | 16   | APB read data bus                                                                |
| <i>xtld_dbg_ref_clk</i>            | 1    | Reference clock for debug only<br>(XpertTeakLite-II clock)                       |
| <i>dctl_sci</i>                    | 54   | Scan inputs                                                                      |
| <i>xtld_glob_wait_r</i>            | 1    | PMU global wait signal to TeakLite-II                                            |
| <i>xtld_haddr_r</i>                | 32   | Scan-out through AHB slave address bus                                           |
| <i>xtld_fmif_faddr_r</i>           | 19   | Scan-out through FLASH write data bus                                            |
| <i>xtld_hsize_r</i>                | 3    | Scan-out through AHB slave transaction size bus                                  |
| <i>xtld_eapb_paddr_r</i>           | 9    | EAPB paddr bus scan mux, inserted in synthesis                                   |
| <i>xtld_eapb_pwrite_r</i>          | 1    | EAPB pwrite scan mux, inserted in synthesis                                      |
| <i>dctl_btdmpr_fsp_oe</i>          | 1    | Receive Channel, Frame Synchronization Pulse<br>Output Enable                    |
| <i>dctl_btdmpr_cntrl_tdmclk_oe</i> | 1    | Receive Channel, TDMCLK Output Enable                                            |
| <i>dctl_btdmpt_plmt_tdout_oe</i>   | 1    | Transmit Channel, Serial Transmit DATA OUT<br>Output Enable                      |
| <i>dctl_btdmpt_fsp_oe</i>          | 1    | Transmit Channel, Frame Synchronization Pulse<br>Output Enable                   |
| <i>dctl_btdmpt_cntrl_tdmclk_oe</i> | 1    | Transmit Channel, TDMCLK Output Enable                                           |
| <i>dctl_ashb_oe</i>                | 1    | AHB Master I/O pad output enable (active low)                                    |
| <i>dctl_eapb_oe</i>                | 1    | APB output enable (active low)                                                   |
| <i>dctl_fmif_pad_oe</i>            | 1    | FMIF data pads output enable                                                     |
| <i>dctl_piu_oe_r</i>               | 1    | APB output enable (active low)                                                   |
| <i>dctl_ashb_ctrl_oe</i>           | 1    | AHB master controls output enable<br>(float the buses when an indication occurs) |
| <i>in_clk</i>                      | 1    | Clock switch output                                                              |
| <i>dctl_boot_vec_r</i>             | 32   | Boot vector                                                                      |
| <i>dctl_debugp_r</i>               | 1    | Debug indication                                                                 |
| <i>dctl_ext_wait_sync_r</i>        | 1    | Synchronized CEVA-TeakLite-II wait-state request                                 |
| <i>dctl_buin0p_sync_r</i>          | 1    | Synchronized user input #0                                                       |
| <i>dctl_buin1p_sync_r</i>          | 1    | Synchronized user input #1                                                       |
| <i>dctl_ext_nmi_sync_r</i>         | 1    | Synchronized external NMI                                                        |
| <i>dctl_ext_int_src_sync_r</i>     | 3    | Synchronized external interrupt                                                  |
| <i>dctl_ext_bp_req_sync_n</i>      | 1    | Synchronized external break point (abort) request                                |
| <i>dctl_pll_lock_sync_r</i>        | 1    | Synchronized PLL lock indication                                                 |

| Output Port Name             | Size | Description                       |
|------------------------------|------|-----------------------------------|
| <i>dctl_pllctrl0_cr</i>      | 16   | PLL configuration register        |
| <i>dctl_pllctrl1_cr</i>      | 16   | PLL configuration register        |
| <i>dctl_pllctrl2_cr</i>      | 16   | PLL configuration register        |
| <i>dctl_pllctrl3_cr</i>      | 16   | PLL configuration register        |
| <i>dctl_prg_bist_in</i>      | 12   | Program BIST input                |
| <i>dctl_timer0_tout_r</i>    | 1    | Program BIST output mux - bit [3] |
| <i>dctl_timer0_pwm_r</i>     | 1    | Program BIST output mux - bit [2] |
| <i>dctl_timer1_tout_r</i>    | 1    | Program BIST output mux - bit [1] |
| <i>dctl_timer1_pwm_r</i>     | 1    | Program BIST output mux - bit [0] |
| <i>dctl_data_bist_in</i>     | 12   | Data BIST input                   |
| <i>dctl_fmif_we_n_r</i>      | 1    | Data BIST output mux              |
| <i>dctl_fmif_oe_n_r</i>      | 1    | Data BIST output mux              |
| <i>dctl_fmif_ce_n_r</i>      | 1    | Data BIST output mux              |
| <i>dctl_fmif_baa_r</i>       | 1    | Data BIST output mux              |
| <i>dctl_L2_bist_in</i>       | 12   | L2 BIST input                     |
| <i>dctl_uout0p</i>           | 1    | L2 BIST output mux                |
| <i>dctl_uout1p</i>           | 1    | L2 BIST output mux                |
| <i>dctl_piu_dsem_int_r</i>   | 1    | L2 BIST output mux                |
| <i>dctl_piu_comrep_int_r</i> | 1    | L2 BIST output mux                |

## 6.3 Monitor Functionality

### 6.3.1 Monitor Counters

The bus monitor mechanism includes a set of 26 counters. The counters are incremented by their corresponding hardware events. All counters are read-only registers. They can be cleared by writing the *CLR* field in the **MONCTRL** register. When written with '1', this bit is automatically cleared by hardware. All counters operate in parallel.

There are 6 events that the counters count:

- Total cycles, number of hclk cycles
- MIM Valid cycles - number of valid transfer cycles issued by each MIM. Cycles in which the *HTRANS* is not IDLE '00' (5 counters)
- MIM received Address not ready. Logic AND between all **AREADY** inputs of each MIM module (5 counters)
- MIM generates data not ready. Counts the number of cycles in which the specific MIM module generate **HREADY** signal low to the corresponding AHB master (5 counters)
- SIM Valid cycle, number of valid transfer cycles. Cycles in which the *HTRANS* is not IDLE '00' (5 counters).
- SIM generates Address not ready. Logic AND between all **AREADY** outputs of each SIM module (5 counters).

Three programmable bits in the **MONCTRL** register control the counters:

- RST – Counters reset. When set to '1' all counters are cleared and event counting begins. This bit is cleared automatically by hardware one cycle after being set.
- EN – Enables counting. This bit must be set in order for the counting to operate

When the **TKL\_TRAPAP** indication is held high (trap interrupt routine) all counter are paused.

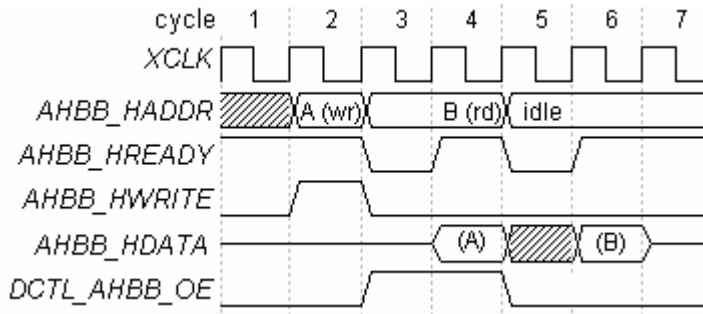
Using the 26 counter the user can have a statistics regarding the percentage of time there was a conflict on a specific AHB slave and the load on a specific AHB master or slave.

### 6.3.2 Bidirectional Data I/O Pad Output Enabling

The I/O pads for each bidirectional data bus (External PIU APB slot, FLASH I/F, APB expansion slot and ASHB bridge) receive an output enable signal from the DEVCNTL module. Output enable signals are active high. Output enable signals of master ports (APB expansion slot, FLASH I/F and ASHB) have similar timing as the write data of their corresponding master. For a slave port (External PIU), the output enable have similar timing as the read data. The example below demonstrates the AHB port data I/O pads output enable behavior:



### 6.3.2.1 AHBB Data Port Output Enabling

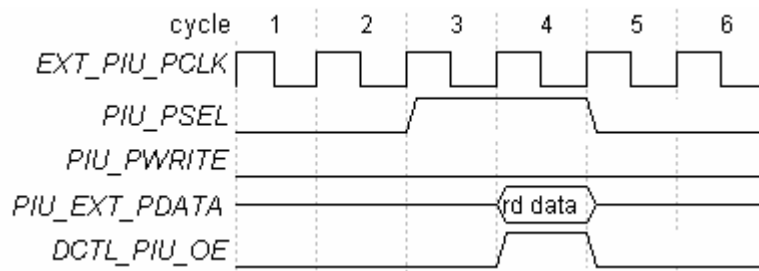


| Signal Name  | Function                         |
|--------------|----------------------------------|
| XCLK         | System clock                     |
| AHBB_HADDR   | AHB address                      |
| AHBB_HREADY  | AHB ready from external slave    |
| AHBB_HWRITE  | AHB write indication             |
| AHBB_HDATA   | Bi-directional AHB data bus      |
| DCTL_AHBB_OE | I/O Pad output enable indication |

**Figure 6-2: AHBB Output Enable Timing**

When both **AHBB\_HREADY** and **AHBB\_HWRITE** are active, the output enable signal, **DCTL\_AHBB\_OE** is cleared (active), when **AHBB\_HREADY** is high and **AHBB\_HWRITE** is low (read transfer), **DCTL\_AHBB\_OE** is set high (inactive).

### 6.3.2.2 PIU External Data Port Output Enabling

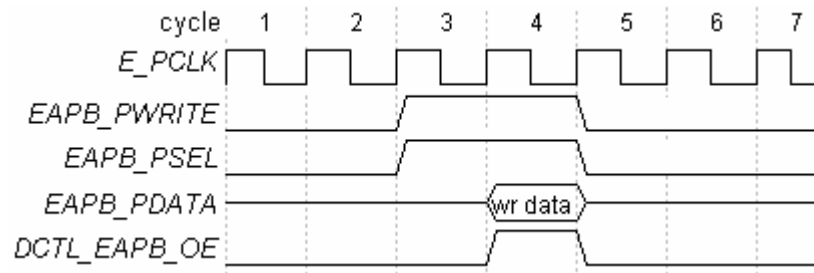


| Signal Name          | Function                                  |
|----------------------|-------------------------------------------|
| <i>EXT_PIU_PCLK</i>  | APB clock from external APB master system |
| <i>PIU_PSEL</i>      | APB controls from external APB master     |
| <i>PIU_PWRITE</i>    | APB controls from external APB master     |
| <i>PIU_EXT_PDATA</i> | Bi-directional APB data bus               |
| <i>DCTL_PIU_OE</i>   | I/O buffer output enable indication       |

**Figure 6-3: PIU Output Enable Timing**

When the XpertTeakLite-II drives output read data over the *PIU\_EXT\_PDATA* bus, the *DCTL\_PIU\_OE* signal is set high to enable the I/O buffer.

### 6.3.2.3 EAPB External Data Port Output Enabling



| Signal Name         | Function                                      |
|---------------------|-----------------------------------------------|
| <i>E_PCLK</i>       | External APB system clock signal (slave side) |
| <i>EAPB_PSEL</i>    | APB controls to external slaves               |
| <i>EAPB_PWRITE</i>  | APB controls to external slaves               |
| <i>EAPB_PDATA</i>   | Bi-directional APB data bus                   |
| <i>DCTL_EAPB_OE</i> | I/O buffer output enable indication           |

**Figure 6-4: EAPB Output Enable Timing**

In this case, *DCTL\_EAPB\_OE* is cleared for a single *E\_PCLK* cycle when (one-hot) when *EAPB\_PWRITE*, *DCTL\_EAPB\_OE* and *EAPB\_PSEL* are high.

## 6.4 Boot Strap Logic

During the reset sequence, XpertTeakLite-II uses a dedicated 32 bit input bus as a boot vector (pointer to the first executable instruction). The DEVCNTRL block generates this boot vector – (**BOOTVECTOR** signal) by using a bi-functional I/O bus (boot-strap), namely *AHBB\_HDATA*. A boot-strap port is captured by the rising edge of the *XTL\_RST\_N* signal. After reset, it retains its original functionality.

## 6.5 System Master Clock Selection

DEVCNTL facilitates a selection between two possible clock signals as the XpertTeakLite-II main clock source:

- The oscillator input clock
- The PLL output clock

The unit assures a glitch free transition, by software, between the two clock sources.

## 6.6 Input Pin Synchronization

The DEVCNTL is responsible for performing a simple synchronization (double latching) of a set of chip inputs. These include:

- *UIA0, IUA1*
- *EXT\_NMI*
- *EXT\_INTSRC[3:0]*
- *EXT\_BP\_REQ*

## 6.7 DEVCNTL Programming Model

DEVCNTL programming registers are mapped to the CEVA-TeakLite-II Z data space. Each register is located at DEVCNTL\_BASE + Offset according to Table 6-3. The value of DEVCNTL\_BASE is specified in Table 4-1. The subfields of each register are described in more detail in the sections that follow.

**Table 6-3: DEVCNTL Configuration Registers**

| Name         | Offset | Description        |        |    |    |    |    |     |   |   |   |          |              |         |        |        |           |           |
|--------------|--------|--------------------|--------|----|----|----|----|-----|---|---|---|----------|--------------|---------|--------|--------|-----------|-----------|
|              |        | 15                 | 14     | 13 | 12 | 11 | 10 | 9   | 8 | 7 | 6 | 5        | 4            | 3       | 2      | 1      | 0         |           |
| PLLCNTL0 *   |        | CP                 |        |    |    |    |    |     |   |   | R |          |              | PLLLOCK | sleep  | PLLrst | byPassPII | byPassCCU |
| PLLCNTL1 *   |        | FBD                |        |    |    |    |    |     |   |   |   | VCOMODE  |              | R       | VCODIV |        |           |           |
| PLLCNTL2 *   |        | R                  |        |    |    |    |    | RES |   |   |   | IPD      |              |         |        |        |           |           |
| PLLCNTL3 *   |        | R                  | OPDMSR |    |    |    |    | OPD |   |   |   | deskewEn | pfdLockDelay |         |        |        |           |           |
| MONCTRL      | 0x04   | Reserved           |        |    |    |    |    |     |   |   |   |          |              |         |        | EN     | CLR       |           |
| CYCLE_L      | 0x05   | CYCLE[15:0]        |        |    |    |    |    |     |   |   |   |          |              |         |        |        |           |           |
| CYCLE_H      | 0x06   | CYCLE[31:16]       |        |    |    |    |    |     |   |   |   |          |              |         |        |        |           |           |
| MIM0_VALID_L | 0x07   | MIM0_VCOUNT[15:0]  |        |    |    |    |    |     |   |   |   |          |              |         |        |        |           |           |
| MIM0_VALID_H | 0x08   | MIM0_VCOUNT[31:16] |        |    |    |    |    |     |   |   |   |          |              |         |        |        |           |           |
| MIM0_WAIT_L  | 0x09   | MIM0_WCOUNT[15:0]  |        |    |    |    |    |     |   |   |   |          |              |         |        |        |           |           |
| MIM0_WAIT_H  | 0x0a   | MIM0_WCOUNT[31:16] |        |    |    |    |    |     |   |   |   |          |              |         |        |        |           |           |
| MIM0_READY_L | 0x0b   | MIM0_DCOUNT[15:0]  |        |    |    |    |    |     |   |   |   |          |              |         |        |        |           |           |
| MIM0_READY_H | 0x0c   | MIM0_DCOUNT[31:16] |        |    |    |    |    |     |   |   |   |          |              |         |        |        |           |           |

| Name         | Offset | Description        |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------------|--------|--------------------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
|              |        | 15                 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| MIM1_VALID_L | 0x0d   | MIM1_VCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM1_VALID_H | 0x0e   | MIM1_VCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM1_WAIT_L  | 0x0f   | MIM1_WCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM1_WAIT_H  | 0x10   | MIM1_WCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM1_READY_L | 0x11   | MIM1_DCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM1_READY_H | 0x12   | MIM1_DCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM2_VALID_L | 0x13   | MIM2_VCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM2_VALID_H | 0x14   | MIM2_VCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM2_WAIT_L  | 0x15   | MIM2_WCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM2_WAIT_H  | 0x16   | MIM2_WCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM2_READY_L | 0x17   | MIM2_DCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM2_READY_H | 0x18   | MIM2_DCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM3_VALID_L | 0x19   | MIM3_VCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM3_VALID_H | 0x1a   | MIM3_VCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM3_WAIT_L  | 0x1b   | MIM3_WCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM3_WAIT_H  | 0x1c   | MIM3_WCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM3_READY_L | 0x1d   | MIM3_DCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM3_READY_H | 0x1e   | MIM3_DCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM4_VALID_L | 0x1f   | MIM4_VCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM4_VALID_H | 0x20   | MIM4_VCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM4_WAIT_L  | 0x21   | MIM4_WCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

| Name         | Offset | Description        |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------------|--------|--------------------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
|              |        | 15                 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| MIM4_WAIT_H  | 0x22   | MIM4_WCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM4_READY_L | 0x23   | MIM4_DCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| MIM4_READY_H | 0x24   | MIM4_DCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM0_VALID_L | 0x25   | SIM0_VCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM0_VALID_H | 0x26   | SIM0_VCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM0_ARB_L   | 0x27   | SIM0_ACOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM0_ARB_H   | 0x28   | SIM0_ACOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM1_VALID_L | 0x29   | SIM1_VCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM1_VALID_H | 0x2a   | SIM1_VCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM1_ARB_L   | 0x2b   | SIM1_ACOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM1_ARB_H   | 0x2c   | SIM1_ACOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM2_VALID_L | 0x2d   | SIM2_VCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM2_VALID_H | 0x2e   | SIM2_VCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM2_ARB_L   | 0x2f   | SIM2_ACOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM2_ARB_H   | 0x30   | SIM2_ACOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM3_VALID_L | 0x31   | SIM3_VCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM3_VALID_H | 0x32   | SIM3_VCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM3_ARB_L   | 0x33   | SIM3_ACOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM3_ARB_H   | 0x34   | SIM3_ACOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM4_VALID_L | 0x35   | SIM4_VCOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM4_VALID_H | 0x36   | SIM4_VCOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| SIM4_ARB_L   | 0x37   | SIM4_ACOUNT[15:0]  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

| Name              | Offset | Description        |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|-------------------|--------|--------------------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
|                   |        | 15                 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|                   |        |                    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| <b>SIM4_ARB_H</b> | 0x38   | SIM4_ACOUNT[31:16] |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| <b>REFCLK_DIV</b> | 0x39   | REFCLK_DIV         |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

\* More details about the PLL configuration fields in registers PLLCNTL[3:0] may be found in the PLL datasheets.

R Reserved bits. These fields are read-only read as zero. If written, the write value must be zero for future compatibility



### 6.7.1 DEVCNTL Registers Field Description

| Register Name | PLLCNTL0            |
|---------------|---------------------|
| Address       | DEVCNTL_BASE + 0x00 |
| Reset Value   | 0x0000              |
| Access        | Read/Write          |

| Field [Bits] | Description                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------|
| CP[15:8]     | charge pump current select                                                                                             |
| PLLLOCK[4]   | Active high PLL Lock indication. When set, indicates the PLL output clock is locked to its desired frequency.          |
| Sleep[3]     | Active high PLL Sleep input pin. When set high, places the PLL in sleep mode                                           |
| PLLrst[2]    | Active high PLL Reset input pin. When set, this input resets the PLL                                                   |
| byPassPII[1] | Active high PLL Bypass pin – in the PLL module. Must be cleared in order to drive PLL output clock as the system clock |
| byPassCCU[0] | Active high PLL Bypass pin – the CCU module. Must be cleared in order to drive PLL output clock as the system clock    |

| Register Name | PLLCNTL1            |
|---------------|---------------------|
| Address       | DEVCNTL_BASE + 0x01 |
| Reset Value   | 0x0000              |
| Access        | Read/Write          |

| Field [Bits]  | Description             |
|---------------|-------------------------|
| FBD[15:6]     | Feedback divider select |
| VCOMODE [5:4] | PLL Mode Select         |
| VCODIV [2:0]  | VCO clock divider       |

| Register Name | PLLCNTL2            |
|---------------|---------------------|
| Address       | DEVCNTL_BASE + 0x02 |
| Reset Value   | 0x0000              |
| Access        | Read/Write          |

| Field [Bits] | Description          |
|--------------|----------------------|
| RES[9:6]     | PLL resistor select  |
| IPD[15:10]   | Input divider select |

|                      |                     |
|----------------------|---------------------|
| <b>Register Name</b> | <b>PLLCNTL3</b>     |
| <b>Address</b>       | DEVCNTL_BASE + 0x03 |
| <b>Reset Value</b>   | 0x0000              |
| <b>Access</b>        | Read/Write          |

| Field [Bits] | Description               |
|--------------|---------------------------|
| OPDMSR       | PLL Duty Cycle definition |
| OPD          | PLL output divider        |
| deskewEn     | Enable Deskew             |
| pdfLockDelay | Lock detect window select |

|                      |                     |
|----------------------|---------------------|
| <b>Register Name</b> | <b>MONCTRL</b>      |
| <b>Address</b>       | DEVCNTL_BASE + 0x04 |
| <b>Reset Value</b>   | 0x0002              |
| <b>Access</b>        | Read/Write          |

| Field [Bits] | Description                                                                                                                                      |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| EN[1]        | Monitor Enable – When set to ‘1’ enables the monitor counters<br>‘1’ – Counting is in progress<br>‘0’ – Monitor counters are paused              |
| CLR[0]       | Monitor Clear – When set to ‘1’ clears all monitor counter. Always read as zero. After being set, this bit is automatically cleared by hardware. |

|                      |                         |
|----------------------|-------------------------|
| <b>Register Name</b> | <b>CYCLE_L/ H</b>       |
| <b>Address</b>       | DEVCNTL_BASE + 0x5/0x06 |
| <b>Reset Value</b>   | 0                       |
| <b>Access</b>        | Read-only               |

| Field [Bits] | Description                                       |
|--------------|---------------------------------------------------|
| CYCLE [31:0] | Cycle Count – 32 bit counter, counts hclk cycles. |

|                      |                          |
|----------------------|--------------------------|
| <b>Register Name</b> | <b>MIM0_VALID_L/ _H</b>  |
| <b>Address</b>       | DEVCNTRL_BASE + 0x7/0x08 |
| <b>Reset Value</b>   | 0                        |
| <b>Access</b>        | Read-only                |

| Field [Bits]          | Description                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| MIM0_VCOUNT<br>[31:0] | MIM 0 Valid Transfer Count – 32 bit counter, counts the cycle in which MIM 0 drives not IDLE transfer. |

|                      |                         |
|----------------------|-------------------------|
| <b>Register Name</b> | <b>MIM0_WAIT_L/ _H</b>  |
| <b>Address</b>       | DEVCNTRL_BASE + 0x9/0xa |
| <b>Reset Value</b>   | 0                       |
| <b>Access</b>        | Read-only               |

| Field [Bits]          | Description                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------|
| MIM0_WCOUNT<br>[31:0] | MIM 0 Wait Count – 32 bit counter, counts the cycle in which MIM 0 received AREADY low. |

|                      |                         |
|----------------------|-------------------------|
| <b>Register Name</b> | <b>MIM0_READY_L/ _H</b> |
| <b>Address</b>       | DEVCNTRL_BASE + 0xb/0xc |
| <b>Reset Value</b>   | 0                       |
| <b>Access</b>        | Read-only               |

| Field [Bits]          | Description                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------|
| MIM0_DCOUNT<br>[31:0] | MIM 0 HREADY Count – 32 bit counter, counts the cycle in which MIM 0 generate HREADY low. |

|                      |                         |
|----------------------|-------------------------|
| <b>Register Name</b> | <b>MIM1_VALID_L/ _H</b> |
| <b>Address</b>       | DEVCNTRL_BASE + 0xd/0xe |
| <b>Reset Value</b>   | 0                       |
| <b>Access</b>        | Read-only               |

| Field [Bits]          | Description                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| MIM1_VCOUNT<br>[31:0] | MIM 1 Valid Transfer Count – 32 bit counter, counts the cycle in which MIM 1 drives not IDLE transfer. |

|                      |                          |
|----------------------|--------------------------|
| <b>Register Name</b> | <b>MIM1_WAIT_L/ _H</b>   |
| <b>Address</b>       | DEVCNTRL_BASE + 0xf/0x10 |
| <b>Reset Value</b>   | 0                        |
| <b>Access</b>        | Read-only                |

| Field [Bits]          | Description                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------|
| MIM1_WCOUNT<br>[31:0] | MIM 1 Wait Count – 32 bit counter, counts the cycle in which MIM 1 received AREADY low. |

|                      |                          |
|----------------------|--------------------------|
| <b>Register Name</b> | <b>MIM1_READY_L/ H</b>   |
| <b>Address</b>       | DEVCNTL_BASE + 0x11/0x12 |
| <b>Reset Value</b>   | 0                        |
| <b>Access</b>        | Read-only                |

| Field [Bits]          | Description                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------|
| MIM1_DCOUNT<br>[31:0] | MIM 1 HREADY Count – 32 bit counter, counts the cycle in which MIM 1 generate HREADY low. |

|                      |                          |
|----------------------|--------------------------|
| <b>Register Name</b> | <b>MIM2_VALID_L/ H</b>   |
| <b>Address</b>       | DEVCNTL_BASE + 0x13/0x14 |
| <b>Reset Value</b>   | 0                        |
| <b>Access</b>        | Read-only                |

| Field [Bits]          | Description                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| MIM2_VCOUNT<br>[31:0] | MIM 2 Valid Transfer Count – 32 bit counter, counts the cycle in which MIM 2 drives not IDLE transfer. |

|                      |                          |
|----------------------|--------------------------|
| <b>Register Name</b> | <b>MIM2_WAIT_L/ H</b>    |
| <b>Address</b>       | DEVCNTL_BASE + 0x15/0x16 |
| <b>Reset Value</b>   | 0                        |
| <b>Access</b>        | Read-only                |

| Field [Bits]          | Description                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------|
| MIM2_WCOUNT<br>[31:0] | MIM 2 Wait Count – 32 bit counter, counts the cycle in which MIM 2 received AREADY low. |

|                      |                          |
|----------------------|--------------------------|
| <b>Register Name</b> | <b>MIM2_READY_L/ H</b>   |
| <b>Address</b>       | DEVCNTL_BASE + 0x17/0x18 |
| <b>Reset Value</b>   | 0                        |
| <b>Access</b>        | Read-only                |

| Field [Bits]          | Description                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------|
| MIM2_DCOUNT<br>[31:0] | MIM 2 HREADY Count – 32 bit counter, counts the cycle in which MIM 2 generate HREADY low. |

|                      |                           |
|----------------------|---------------------------|
| <b>Register Name</b> | <b>MIM3_VALID_L/ _H</b>   |
| <b>Address</b>       | DEVCNTRL_BASE + 0x19/0x1a |
| <b>Reset Value</b>   | 0                         |
| <b>Access</b>        | Read-only                 |

| Field [Bits]          | Description                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| MIM3_VCOUNT<br>[31:0] | MIM 3 Valid Transfer Count – 32 bit counter, counts the cycle in which MIM 3 drives not IDLE transfer. |

|                      |                           |
|----------------------|---------------------------|
| <b>Register Name</b> | <b>MIM3_WAIT_L/ _H</b>    |
| <b>Address</b>       | DEVCNTRL_BASE + 0x1b/0x1c |
| <b>Reset Value</b>   | 0                         |
| <b>Access</b>        | Read-only                 |

| Field [Bits]          | Description                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------|
| MIM3_WCOUNT<br>[31:0] | MIM 3 Wait Count – 32 bit counter, counts the cycle in which MIM 3 received AREADY low. |

|                      |                           |
|----------------------|---------------------------|
| <b>Register Name</b> | <b>MIM3_READ_L/ _H Y</b>  |
| <b>Address</b>       | DEVCNTRL_BASE + 0x1d/0x1e |
| <b>Reset Value</b>   | 0                         |
| <b>Access</b>        | Read-only                 |

| Field [Bits]          | Description                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------|
| MIM3_DCOUNT<br>[31:0] | MIM 3 HREADY Count – 32 bit counter, counts the cycle in which MIM 3 generate HREADY low. |

|                      |                           |
|----------------------|---------------------------|
| <b>Register Name</b> | <b>MIM4_VALID_L/ _H</b>   |
| <b>Address</b>       | DEVCNTRL_BASE + 0x1f/0x20 |
| <b>Reset Value</b>   | 0                         |
| <b>Access</b>        | Read-only                 |

| Field [Bits]          | Description                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| MIM4_VCOUNT<br>[31:0] | MIM 4 Valid Transfer Count – 32 bit counter, counts the cycle in which MIM 4 drives not IDLE transfer. |

|                      |                           |
|----------------------|---------------------------|
| <b>Register Name</b> | <b>MIM4_WAIT_L/ _H</b>    |
| <b>Address</b>       | DEVCNTRL_BASE + 0x21/0x22 |
| <b>Reset Value</b>   | 0                         |
| <b>Access</b>        | Read-only                 |

| Field [Bits]          | Description                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------|
| MIM4_WCOUNT<br>[31:0] | MIM 4 Wait Count – 32 bit counter, counts the cycle in which MIM 4 received AREADY low. |

|                      |                          |
|----------------------|--------------------------|
| <b>Register Name</b> | <b>MIM4_READY_L/_H</b>   |
| <b>Address</b>       | DEVCNTL_BASE + 0x23/0x24 |
| <b>Reset Value</b>   | 0                        |
| <b>Access</b>        | Read-only                |

| Field [Bits]          | Description                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------|
| MIM4_DCOUNT<br>[31:0] | MIM 4 HREADY Count – 32 bit counter, counts the cycle in which MIM 4 generate HREADY low. |

|                      |                           |
|----------------------|---------------------------|
| <b>Register Name</b> | <b>SIM0_VALID_L/ H</b>    |
| <b>Address</b>       | DEVCNTRL_BASE + 0x25/0x26 |
| <b>Reset Value</b>   | 0                         |
| <b>Access</b>        | Read-only                 |

| Field [Bits]          | Description                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------|
| SIM0_VCOUNT<br>[31:0] | SIM 0 Valid Transfer Count – 32 bit counter, counts the cycle in which SIM 0 drives a valid transfer to the AHB slave. |

|                      |                           |
|----------------------|---------------------------|
| <b>Register Name</b> | <b>SIM0_ARB_L/ H</b>      |
| <b>Address</b>       | DEVCNTRL_BASE + 0x27/0x28 |
| <b>Reset Value</b>   | 0                         |
| <b>Access</b>        | Read-only                 |

| Field [Bits]          | Description                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| SIM0_ACOUNT<br>[31:0] | SIM 0 AREADY Count – 32 bit counter, counts the cycle in which SIM 0 drives AREADY low. Number of cycles of contention between two masters. |

|                      |                           |
|----------------------|---------------------------|
| <b>Register Name</b> | <b>SIM1_VALID_L/ H</b>    |
| <b>Address</b>       | DEVCNTRL_BASE + 0x29/0x2a |
| <b>Reset Value</b>   | 0                         |
| <b>Access</b>        | Read-only                 |

| Field [Bits]          | Description                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------|
| SIM1_VCOUNT<br>[31:0] | SIM 1 Valid Transfer Count – 32 bit counter, counts the cycle in which SIM 1 drives a valid transfer to the AHB slave. |

|                      |                          |
|----------------------|--------------------------|
| <b>Register Name</b> | <b>SIM1_ARB_L/ H</b>     |
| <b>Address</b>       | DEVCNTL_BASE + 0x2b/0x2c |
| <b>Reset Value</b>   | 0                        |
| <b>Access</b>        | Read-only                |

| Field [Bits]          | Description                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| SIM1_ACOUNT<br>[31:0] | SIM 1 AREADY Count – 32 bit counter, counts the cycle in which SIM 1 drives AREADY low. Number of cycles of contention between two masters. |

|                      |                          |
|----------------------|--------------------------|
| <b>Register Name</b> | <b>SIM2_VALID_L/ H</b>   |
| <b>Address</b>       | DEVCNTL_BASE + 0x2d/0x2e |
| <b>Reset Value</b>   | 0                        |
| <b>Access</b>        | Read-only                |

| Field [Bits]          | Description                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------|
| SIM2_VCOUNT<br>[31:0] | SIM 2 Valid Transfer Count – 32 bit counter, counts the cycle in which SIM 2 drives a valid transfer to the AHB slave. |

|                      |                          |
|----------------------|--------------------------|
| <b>Register Name</b> | <b>SIM2_ARB_L/ H</b>     |
| <b>Address</b>       | DEVCNTL_BASE + 0x2f/0x30 |
| <b>Reset Value</b>   | 0                        |
| <b>Access</b>        | Read-only                |

| Field [Bits]          | Description                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| SIM2_ACOUNT<br>[31:0] | SIM 2 AREADY Count – 32 bit counter, counts the cycle in which SIM 2 drives AREADY low. Number of cycles of contention between two masters. |

|                      |                          |
|----------------------|--------------------------|
| <b>Register Name</b> | <b>SIM3_VALID_L/ H</b>   |
| <b>Address</b>       | DEVCNTL_BASE + 0x31/0x32 |
| <b>Reset Value</b>   | 0                        |
| <b>Access</b>        | Read-only                |

| Field [Bits]          | Description                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------|
| SIM3_VCOUNT<br>[31:0] | SIM 3 Valid Transfer Count – 32 bit counter, counts the cycle in which SIM 3 drives a valid transfer to the AHB slave. |



|                      |                           |
|----------------------|---------------------------|
| <b>Register Name</b> | <b>SIM3_ARB_L/ H</b>      |
| <b>Address</b>       | DEVCNTRL_BASE + 0x33/0x34 |
| <b>Reset Value</b>   | 0                         |
| <b>Access</b>        | Read-only                 |

| Field [Bits]          | Description                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| SIM3_ACOUNT<br>[31:0] | SIM 3 AREADY Count – 32 bit counter, counts the cycle in which SIM 3 drives AREADY low. Number of cycles of contention between two masters. |

|                      |                           |
|----------------------|---------------------------|
| <b>Register Name</b> | <b>SIM4_VALID_L/ H</b>    |
| <b>Address</b>       | DEVCNTRL_BASE + 0x35/0x36 |
| <b>Reset Value</b>   | 0                         |
| <b>Access</b>        | Read-only                 |

| Field [Bits]          | Description                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------|
| SIM4_VCOUNT<br>[31:0] | SIM 4 Valid Transfer Count – 32 bit counter, counts the cycle in which SIM 4 drives a valid transfer to the AHB slave. |

|                      |                           |
|----------------------|---------------------------|
| <b>Register Name</b> | <b>SIM4_ARB_L/ H</b>      |
| <b>Address</b>       | DEVCNTRL_BASE + 0x37/0x38 |
| <b>Reset Value</b>   | 0                         |
| <b>Access</b>        | Read-only                 |

| Field [Bits]          | Description                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| SIM4_ACOUNT<br>[31:0] | SIM 4 AREADY Count – 32 bit counter, counts the cycle in which SIM 4 drives AREADY low. Number of cycles of contention between two masters. |

|                      |                      |
|----------------------|----------------------|
| <b>Register Name</b> | <b>REFCLK_DIV</b>    |
| <b>Address</b>       | DEVCNTRL_BASE + 0x39 |
| <b>Reset Value</b>   | 0                    |
| <b>Access</b>        | Read/Write           |

| Field [Bits]         | Description                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| REFCLK_DIV<br>[15:0] | SIM 4 AREADY Count – 32 bit counter, counts the cycle in which SIM 4 drives AREADY low. Number of cycles of contention between two masters. |

## 7 Input/Output Ports

Table 7-1 lists the functional I/O pins of the XpertTeakLite-II Dev chip:

**Table 7-1: I/O Pins**

| Port Name                      | Dir    | Size | Description                                            |
|--------------------------------|--------|------|--------------------------------------------------------|
| <b>AHB Master port</b>         |        |      |                                                        |
| <i>ashb_hz_enforce_u</i>       | input  | 1    | Enforces a high-Z state on all AHB Master output ports |
| <i>s_hready_u</i>              | input  | 1    | AHB Slave ready indication                             |
| <i>s_hresp_u</i>               | input  | 1    | AHB Slave response                                     |
| <i>mst_hdata_b</i>             | io     | 32   | Master port read/write data                            |
| <i>s_hclk_u</i>                | input  | 1    | AHB clock driven from external AHB system (slave side) |
| <i>xtld_haddr_r_u</i>          | output | 32   | Address bus to the slave                               |
| <i>xtld_hsize_r_u</i>          | output | 3    | Transaction size to the slave                          |
| <i>xtld_hburst_r_u</i>         | output | 3    | Burst type to the slave                                |
| <i>xtld_htrans_r_u</i>         | output | 2    | Transaction code to the slave                          |
| <i>xtld_hwrite_r_u</i>         | output | 1    | Write transaction to the slave                         |
| <b>PIU</b>                     |        |      |                                                        |
| <i>xtld_piu_dsem_int_r_u</i>   | output | 1    | Semaphore interrupt to External MCU                    |
| <i>xtld_piu_comrep_int_r_u</i> | output | 1    | Command read interrupt to External MCU                 |
| <i>xtld_piu_wr_err_int_r_u</i> | output | 1    | External MCU write error indication                    |
| <i>ext_pdata_b</i>             | io     | 16   | APB data to/from external APB master host              |
| <i>ext_pclk_u</i>              | input  | 1    | Clock input from external APB master                   |
| <i>ext_pwrite_u</i>            | input  | 1    | APB write from external APB master host                |
| <i>ext_paddr_u</i>             | input  | 4    | APB address from external APB master host              |
| <i>ext_psel_u</i>              | input  | 1    | APB select from external APB master host               |
| <i>ext_penable_u</i>           | input  | 1    | APB enable from external APB master host               |
| <b>MAC</b>                     |        |      |                                                        |
| <i>col_u</i>                   | input  | 1    | Collision detect signal from the PHY                   |
| <i>crs_u</i>                   | input  | 1    | Carrier sense signal from the PHY                      |
| <i>rx_d_u</i>                  | input  | 4    | Transmit data to the PHY                               |
| <i>rx_er_u</i>                 | input  | 1    | Receive error signal from the PHY                      |
| <i>rx_dv_u</i>                 | input  | 1    | Receive data valid signal from the PHY                 |
| <i>link_u</i>                  | input  | 1    | Status of link pin from the PHY                        |
| <i>ref_clk_u</i>               | input  | 1    | 50MHz reference clock input                            |
| <i>rx_clk_source_u</i>         | input  | 1    | MAC Receive data clock from MII                        |
| <i>tx_clk_source_u</i>         | input  | 1    | MAC Transmit clock source                              |
| <i>xtld_mac_tx_er_u</i>        | output | 1    | Transmit error signal to the PHY                       |
| <i>xtld_mac_txd_u</i>          | output | 4    | Transmit data to the PHY                               |
| <i>xtld_mac_tx_en_u</i>        | output | 1    | Transmit enable signal to the PHY                      |
| <i>xtld_mac_mdc_u</i>          | output | 1    | Management data clock                                  |
| <i>mac_mdio_b</i>              | io     | 1    | Management data input/output                           |

| Port Name                      | Dir    | Size | Description                                                                                                                                                         |
|--------------------------------|--------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ICU</b>                     |        |      |                                                                                                                                                                     |
| <i>ext_int_src_u</i>           | input  | 3    | External interrupts to ICU module                                                                                                                                   |
| <i>ext_nmi_u</i>               | input  | 1    | External NMI interrupt to ICU module                                                                                                                                |
| <b>GPIO</b>                    |        |      |                                                                                                                                                                     |
| <i>gpio_data_b</i>             | io     | 8    | General Purpose input/output lines                                                                                                                                  |
| <b>TDM Rx/Tx</b>               |        |      |                                                                                                                                                                     |
| <i>tdm_rx_clk_b</i>            | io     | 1    | BTDMRx 0 TDM clock                                                                                                                                                  |
| <i>tdm_rx_fsp_b</i>            | io     | 1    | BTDMRx 0 FSP line                                                                                                                                                   |
| <i>tdm_tx_clk_b</i>            | io     | 1    | TDM clock for Tx 0                                                                                                                                                  |
| <i>tdm_tx_fsp_b</i>            | io     | 1    | TDM FSP for Tx 0                                                                                                                                                    |
| <i>ext_tdm_rd_in_u</i>         | input  | 1    | BTDM Rx Data input line                                                                                                                                             |
| <i>xtld_btldmpt_tdout_cr_u</i> | output | 1    | Transmit Channel 0 - Serial Transmit DATA OUT                                                                                                                       |
| <b>PMU and PLL</b>             |        |      |                                                                                                                                                                     |
| <i>ex_wait_u</i>               | input  | 1    | External CEVA-TeakLite-II wait-state request                                                                                                                        |
| <i>ex_wake_u</i>               | input  | 1    | External clock recovery request                                                                                                                                     |
| <i>ex_rst_n_u</i>              | input  | 1    | Chip asynchronous reset signal                                                                                                                                      |
| <i>xtld_rst_n_u</i>            | output | 1    | PMU reset to XpertTeakLite-II (Includes OCEM reset)                                                                                                                 |
| <i>clki_u</i>                  | input  | 1    | Clock input from oscillator to on-chip PLL                                                                                                                          |
| <i>xtld_glob_wait_r_u</i>      | output | 1    | PMU global wait signal of TeakLite-II clock signal.<br>This signal is output for observability only)<br>(this is the internal clock gate signal delayed by 1 cycle) |
| <b>Timers</b>                  |        |      |                                                                                                                                                                     |
| <i>xtld_timer0_tout_r_u</i>    | output | 1    | Timer_unit 0 interrupt                                                                                                                                              |
| <i>xtld_timer0_pwm_r_u</i>     | output | 1    | Timer_unit 0 PWM output signal                                                                                                                                      |
| <i>xtld_timer1_tout_r_u</i>    | output | 1    | Timer_unit 1 interrupt                                                                                                                                              |
| <i>ext_clktimer0_u</i>         | input  | 1    | External Clock for TIMER_UNIT0                                                                                                                                      |
| <i>ext_clktimer1_u</i>         | input  | 1    | External Clock for TIMER_UNIT1                                                                                                                                      |
| <i>xtld_timer1_pwm_r_u</i>     | output | 1    | Timer_unit 1 PWM output signal                                                                                                                                      |
| <b>EAPB (APB Expansion)</b>    |        |      |                                                                                                                                                                     |
| <i>xtld_eapb_pwrite_r_u</i>    | output | 1    | Write signal towards EAPB                                                                                                                                           |
| <i>xtld_eapb_psel_r_u</i>      | output | 1    | Select signal towards EAPB                                                                                                                                          |
| <i>xtld_eapb_penable_r_u</i>   | output | 1    | Select signal towards EAPB                                                                                                                                          |
| <i>xtld_eapb_paddr_r_u</i>     | output | 9    | Address bus towards EAPB                                                                                                                                            |
| <i>eapb_pclk_u</i>             | input  | 1    | External APB clock                                                                                                                                                  |
| <i>eapb_pdata_b</i>            | io     | 16   | APB read/write data bus                                                                                                                                             |
| <b>FMIF (Flash Interface)</b>  |        |      |                                                                                                                                                                     |
| <i>fpause_u</i>                | input  | 1    | Pause transfer indication                                                                                                                                           |
| <i>fstat_u</i>                 | input  | 1    | General purpose status input                                                                                                                                        |
| <i>xtld_fmif_faddr_r_u</i>     | output | 19   | FLASH address bus                                                                                                                                                   |
| <i>xtld_fmif_we_n_r_u</i>      | output | 1    | Write enable - active low                                                                                                                                           |
| <i>xtld_fmif_oe_n_r_u</i>      | output | 1    | Output enable - active low                                                                                                                                          |
| <i>xtld_fmif_ce_n_r_u</i>      | output | 1    | Chip enable - active low                                                                                                                                            |

| Port Name                  | Dir    | Size | Description                          |
|----------------------------|--------|------|--------------------------------------|
| <i>xtld_fmif_baa_r_u</i>   | output | 1    | Burst address advance indication     |
| <i>xtld_fmif_lba_r_u</i>   | output | 1    | Latch first burst address            |
| <i>xtld_fmif_fclk_r_u</i>  | output | 1    | Burst mode flash clock signal        |
| <i>fdata_b</i>             | io     | 16   | Memory read/write data bus           |
| <i>xtld_fmif_rst_n_u</i>   | output | 1    | Active low reset signal to the flash |
| <b>JTAG, Test and Misc</b> |        |      |                                      |
| <i>otck_u</i>              | input  | 1    | JTAG TCK                             |
| <i>otdi_u</i>              | input  | 1    | JTAG TDI                             |
| <i>xtld_jtdo_u</i>         | output | 1    | JTAG TDO                             |
| <i>otms_u</i>              | input  | 1    | JTAG TMS                             |
| <i>scanen_u</i>            | input  | 1    | Scan mode indication                 |
| <i>testmodep_u</i>         | input  | 1    | Test mode indication                 |
| <i>ext_bp_req_u</i>        | input  | 1    | External abort (breakpoint) request  |
| <i>buin0p_u</i>            | input  | 1    | CEVA-TeakLite-II User input 0        |
| <i>buin1p_u</i>            | input  | 1    | CEVA-TeakLite-II User input 1        |
| <i>xtld_dbg_ref_clk_u</i>  | output | 1    | Reference clock for debug only       |
| <b>BIST</b>                |        |      |                                      |
| <i>bist_en_u</i>           | input  | 1    | BIST mode                            |
| <i>xtld_ptrapap_u</i>      | output | 1    | Trap routine execution               |
| <i>xtld_uout0p_u</i>       | output | 1    | User output 1                        |
| <i>xtld_uout1p_u</i>       | output | 1    | User output 0                        |